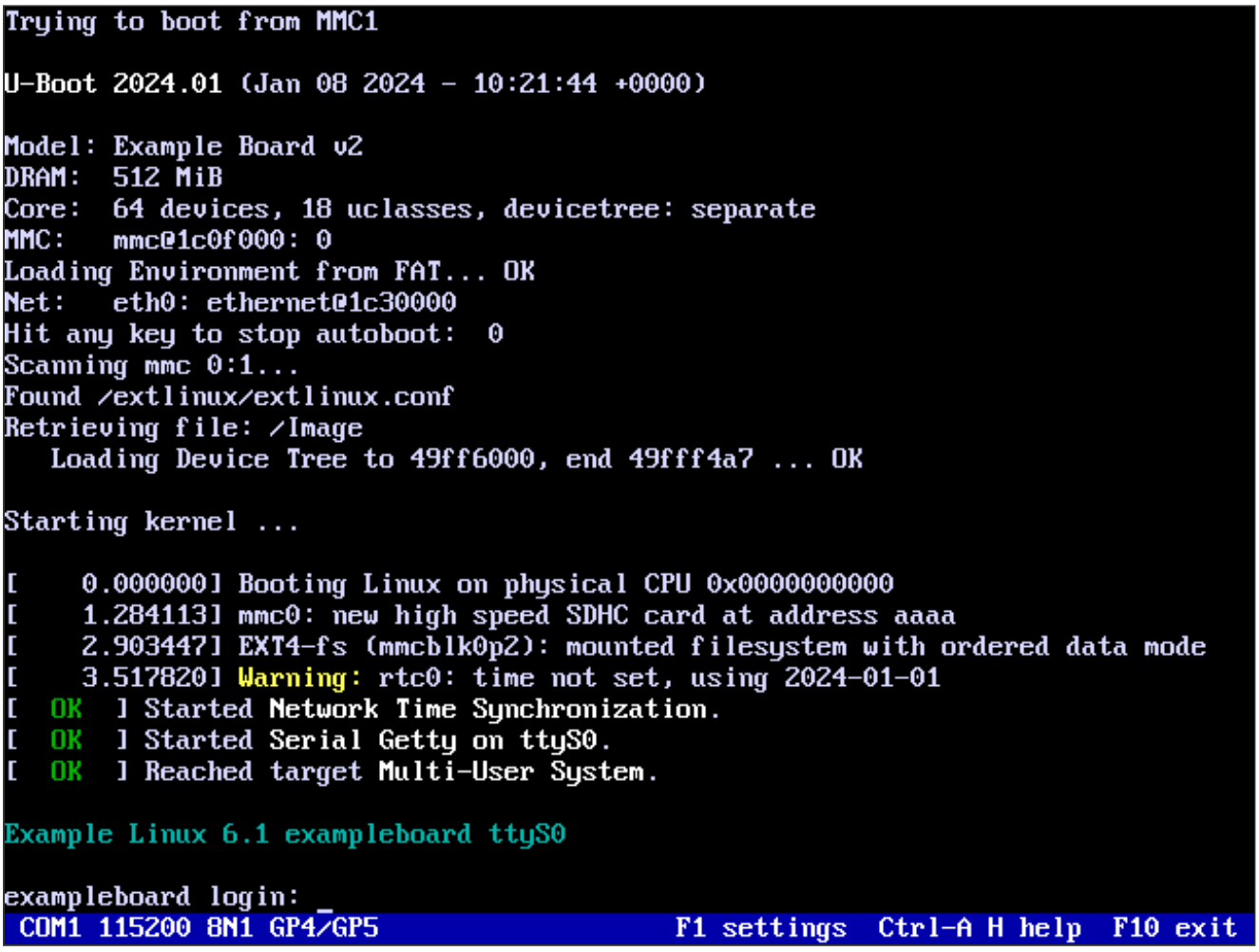
Pico 2 mini OS Serial Terminal

Pico 2 mini OS · October 31, 2026



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`term [baud] [format]` (or `serial`) is a terminal on the tty1 serial port (BASIC's COM1), or on the debug port GP0/GP1 while the serial console is off, with a settings box, logging and file transfer by XMODEM, YMODEM and ZMODEM.

The terminal

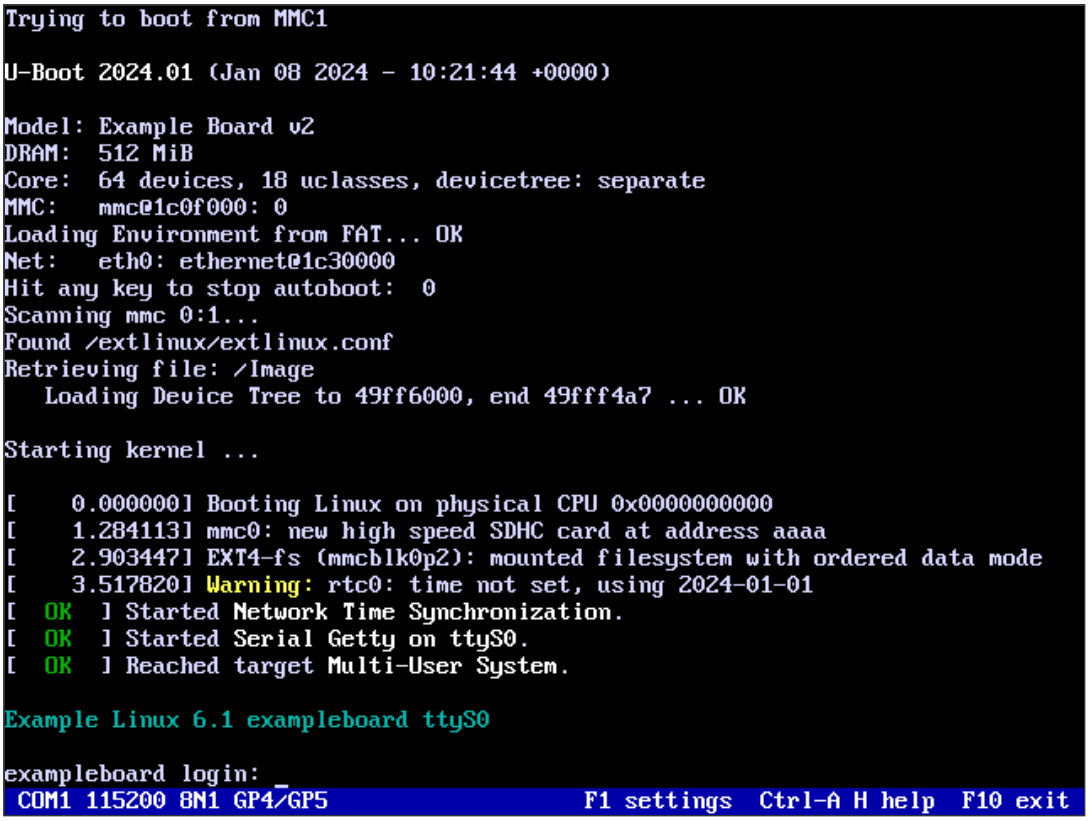
`term` (or `serial`) is a terminal on tty1, the serial port BASIC calls COM1. It is UART1, TX on GP4 and RX on GP5 unless the settings say otherwise (3.3 V levels; connect the far end's RX to TX, TX to RX, and ground).

The debug port (UART0, TX GP0 / RX GP1) can be the terminal's port too, shown as tty0, while the serial console is off (`sercon off`, as at start-up): choose it under **Pins** in the settings. The terminal then has that port to itself at its own speed and format; the debug messages wait, and it goes back to 115200 8N1 when you leave. While the serial console is on (`sercon on`), GP0/GP1 isn't offered, and a saved setting for it says so instead of starting. There is no RTS/CTS on GP0/GP1.

```
term                the saved settings (115200 8N1 at first)
term 9600           another speed this time
term 9600 7E1       speed and format
```

What arrives goes through the console's VT100/ANSI handling, so colours and cursor movement from the far end work. As in the text editor, row 1 is a menu bar, rows 2-29 are the terminal (set the far end to 28 rows, e.g. `stty rows 28`; its cursor moves and screen clears stay inside them) and row 30 is a status line with the port settings. What arrives also goes to the debug serial port when the serial console is on (`sercon`).

Menus: click **File** (Send text file, Log to file, Exit), **Transfer** (Upload, Download), **Settings** (Port settings, Local echo, Send BREAK, Clear screen) or **Help** (Keys, About) on the menu bar, or press F9 (or Ctrl-A M), or Alt and a title's first letter (Alt-F, Alt-T, Alt-S, Alt-H). Left / Right go to the next menu (or move the pointer onto its title), Up / Down and Enter choose, Esc closes; you can also press on a title, move onto an item and let go. The right mouse button opens a menu with all of these. Each item shows its key. The coloured letter in a title (Alt and it) or an item (press it while the menu is open) is its key letter. While a menu is open, what arrives waits and then shows when it closes.



Above: a board's boot messages arriving on tty1 at 115200 baud, colours and all. (The sample session is made up; any device with a serial console looks much the same.)

Key	Action
F1, or Ctrl-A S	Settings box
F10, or Ctrl-A X	Leave the terminal
F9, or Ctrl-A M	The menus
Alt-F, Alt-T, Alt-S, Alt-H	The File, Transfer, Settings or Help menu (other Alt keys go to the far end)
Ctrl-A B	Send a BREAK (250 ms)
Ctrl-A C	Clear the screen
Ctrl-A E	Local echo on / off
Ctrl-A L	Log what arrives to a file (default `TERM.LOG`, added to); again to stop
Ctrl-A F	Send a file as it is (Esc stops)
Ctrl-A U	Upload: send files with XMODEM, XMODEM-1K, YMODEM or ZMODEM
Ctrl-A D	Download: receive files with XMODEM, YMODEM or ZMODEM
Ctrl-A H	Key help page
Ctrl-A Ctrl-A	Send Ctrl-A

Other keys go to the far end; arrows, Home, End, Delete and function keys as VT100 sequences.

The settings box opens over the terminal (Up/Down or Tab choose, and go on to the buttons, where Left/Right move and Enter presses; Left/Right or Space change a setting; with the mouse, click a setting's `<` or `>`, or turn the wheel over it):

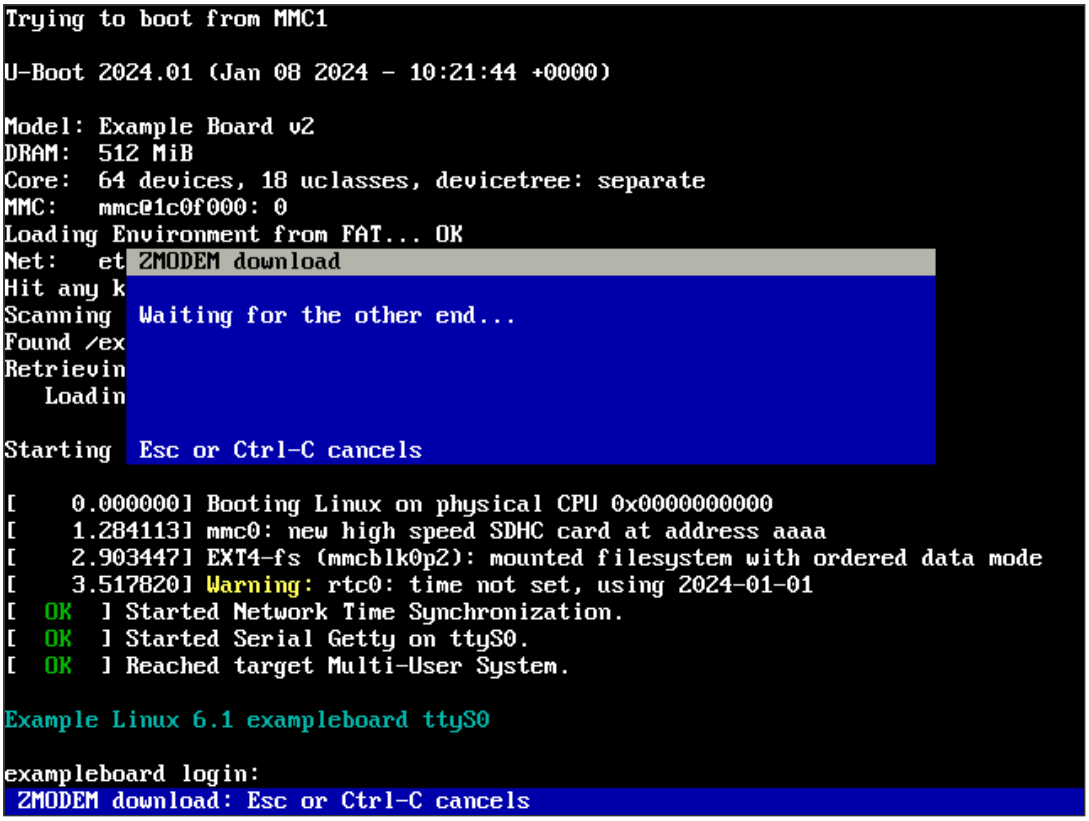
Setting	Choices
Baud rate	300 to 921600 from a list, or Enter to type any rate from 50 to 4000000
Data bits	5, 6, 7, 8
Parity	None, Even, Odd
Stop bits	1, 2
Pins	TX/RX on GP0/GP1 (the debug port, tty0; only while `sercon` is off), GP4/GP5, GP8/GP9, GP20/GP21 or GP24/GP25
Flow control	None, or RTS/CTS on the next two pins (CTS GP6 / RTS GP7 for GP4/GP5, and so on; not on GP0/GP1)
Local echo	Off, On
Enter sends	CR, LF, or CR LF
Received CR	as it is, or with a line feed added
Backspace sends	DEL (127) or BS (8)
Upload folder	where the names typed in the upload box are looked for (a name starting with `/` is used as it is)
Download folder	where received files are saved

OK (Esc, Enter or `O`) uses them, **Save as default** (`S`) also saves them in `/TERM.CFG`, and **Cancel** (`Q`, or the right button) drops the changes. Enter on the baud rate types one. Changing the port settings reopens the port.

File transfer: XMODEM, YMODEM, ZMODEM

Ctrl-A U (or Transfer > Upload) sends files and Ctrl-A D (Transfer > Download) receives them. A box opens: choose the protocol with Up/Down or a click (it remembers the last one; ZMODEM at first), type the files to send (or, for an XMODEM download, the name to save as), then Start (Enter) or Cancel (Esc, or the right button). **Browse...** (or Ctrl-O) in the upload box opens the file selector in the upload folder; the file chosen is added to the list.

The file selector lists a folder, folders first. Enter or a double-click opens a folder (`..` and Backspace go up) or chooses a file; a letter jumps to the next name starting with it; the wheel scrolls; Tab goes to the buttons. For the folder settings (Enter or a click on one in the settings box), **Use this folder** takes the folder shown at the top. Both folders are kept with the other settings when you press Save as default; empty means the current folder. A window shows the file, bytes and percentage, speed, retries and files done; Esc or Ctrl-C cancels (the other end is told, with CAN characters).



Above: Ctrl-A D with ZMODEM: the window waiting for the far end to start sending (`sz file` there). Once a file starts, the window shows its name, bytes, speed and retries.

Protocol	Sending	Receiving
XMODEM	one file, 128-byte blocks, CRC-16 or checksum (the receiver chooses)	one file; you type the name to save it as. Its size is rounded up to 128 bytes, padded with Ctrl-Z, as XMODEM has no sizes
XMODEM-1K	one file, 1024-byte blocks	(the same as XMODEM)
YMODEM	several files with names and sizes	the sender's names; exact sizes
ZMODEM	several files, streaming	the sender's names; carries on with a partial file when the sender asks (`sz -r`)

- Files to send: type one or more names separated by spaces; `*` and `?` work in the last part, e.g. `EXAMPLES/*.BAS` (up to 32 files).
- Received files go in the download folder (the current folder unless set). A name that is taken gets `-1`, `-2` ... before the extension, so nothing is overwritten.
- A ZMODEM download starts by itself when the far end runs `sz file`: the terminal sees the start of ZMODEM and opens the window.
- ZMODEM uploads first send `rz` and Enter, which starts the receiver on a Unix shell.
- A broken XMODEM or YMODEM download is deleted; a broken ZMODEM one is kept so that `sz -r` can finish it.
- The receiver asks a ZMODEM sender to wait for an ACK every 4 KB, so a slow drive never loses data; RTS/CTS flow control also works.